

Alan Hu . Jarvis Consulting

Recent McGill University graduate in Statistics and Computer Science with experience in machine learning, data engineering, analytics, and software development. Skilled in building end-to-end ML and data pipelines using Python, PySpark, SQL, Databricks, Azure, and modern data architectures. Experienced in developing data-driven solutions through model development, feature engineering, and scalable data workflows, with a strong foundation in statistical analysis and software engineering practices. Looking to solve complex problems and deliver impactful solutions through strong technical communication and collaboration.

Skills

Proficient: Linux/Bash, RDBMS/SQL, Agile/Scrum, Git, Python (Pandas, Sci-kit), ETL Pipelines

Competent: ML concepts, Java, Databricks, Maven, Docker, Apache Spark/PySpark, Apache Airflow

Familiar: NoSQL/MongoDB, GCP/BigQuery, AWS, Snowflake, Power BI

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_AlanHu/

Credit Risk Prediction [GitHub]: Built a predictive credit risk modeling pipeline to identify high-risk loan applicants and reduce potential defaults using Python and Scikit-learn. Implemented automated preprocessing with feature engineering, missing-value handling, encoding, and scaling, followed by comparative evaluation of multiple classification algorithms. Optimized model performance through hyperparameter tuning, cross-validation, and imbalance handling techniques, while leveraging SHAP-based explainability to analyze feature contributions and improve model transparency.

Cluster Monitor [GitHub]: Automated Linux cluster analysis (LCA) and node orchestration using Bash scripting and PostgreSQL to centralize and analyze cluster resource metrics. Containerized data collection and reporting workflows with Docker and managed version control using Git to ensure reproducibility and collaborative development. Enabled scalable SQL-based reporting on node usage trends to support data-driven capacity planning and resource allocation.

Core Java Apps [GitHub]:

- Twitter App: Placeholder to allow profile_app.sh to work.
- JDBC App: Placeholder to allow profile_app.sh to work.
- Grep App: Built a Java-based grep application to recursively scan large directory structures, apply regex-based filtering, and persist matched results to disk with efficient I/O handling. Implemented modular, interface-driven design with robust exception handling, UTF-8 encoding, and stream-based file traversal using Java NIO. Managed dependencies and project lifecycle using Maven, and integrated structured logging with SLF4J and Log4j for observability.

Python Data Analytics [GitHub]: Spearheaded a Python-based data analysis PoC for a retail gift shop to diagnose stagnant revenue performance. Used Pandas, NumPy, Jupyter Notebook, and SQL to perform exploratory analysis, customer segmentation, and trend analysis, delivering targeted recommendations aligned with the shop's customer base to drive higher engagement and incremental sales.

Highlighted Projects

Databricks Bullhorn CRM Pipeline: Maintained and enhanced enterprise ETL pipelines in Databricks using PySpark, Delta Live Tables, and Medallion Architecture (Bronze, Silver, Gold) to ingest, transform, and curate CRM data for downstream reporting and analytics. Developed REST API-based ingestion workflows using token authentication, orchestrated tier-specific pipelines through centralized Databricks Jobs, and implemented testing pipelines and validation checks to ensure data quality, code reliability, and production readiness.

E-commerce Data Platform Architecture [GitHub]: Developed a hybrid on-premise/cloud data platform for an e-commerce company, integrating MySQL for OLTP and MongoDB for NoSQL storage. Built automated ETL pipelines using Python and Apache Airflow to load data into a PostgreSQL staging warehouse and Hadoop for large-scale analytics. Applied PySpark and SparkML to support big data processing and sales forecasting within the analytics pipeline. Enabled business intelligence reporting and KPI dashboards through IBM Cognos Analytics to support data-driven decision-making.

Professional Experiences

Data Engineer/Jr Technical Instructor, Jarvis (Nov 2025 - Present): Developed machine learning and data engineering solutions using Python, Pandas, NumPy, and Azure Databricks, building scalable data pipelines and analytics workflows to support business insights. Designed and implemented ML workflows including data preprocessing, feature engineering, model evaluation, and automated analysis while leveraging Linux, PostgreSQL, and cloud-based environments. Applied software engineering practices including clean architecture, unit testing, Docker-based deployment, Git version control, and technical documentation to improve maintainability and reproducibility. Supported technical training initiatives by delivering instructional modules, developing assessments, and mentoring junior developers through engineering and machine learning concepts.

Education

McGill University (Sep 2021 - Apr 2025), Bachelor of Science, Statistics and Computer Science

Miscellaneous

- Google Advanced Data Analytics Certificate (July 2025)
- IBM Data Engineering Certificate (October 2025)
- Experience with tutoring and swim instructing
- Passion for video editing and table tennis